

PART-2



How to choose a Research Problem?

What to Consider When Choosing a Problem

- ❧ Is the problem of a high significance?
- ❧ Is the problem considered a hot topic for many researchers?
- ❧ Is the problem valid?
- ❧ Did anyone have solved this problem before?
 - ❧ If the answer to 4 is “Yes”, then, How many of them? Many or little?
 - ❧ If the answer to 4 is “Yes”, then, Is there any of them have solved the problem completely with very reasonable costs and very satisfying results?
 - ❧ If the answer to 4 is “Yes”, then, How competitive your solution can be compares to theirs?



How to form a Problem Statement?

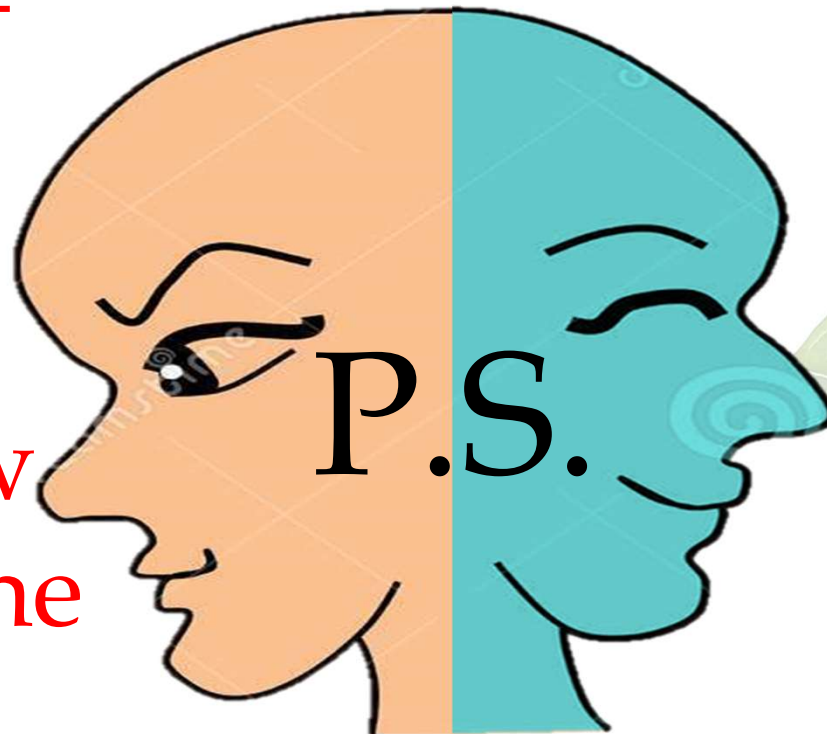
Problem Statement (P.S.) Faces



Technical
Cause

Real-life
Effect

Do not
know how
to divide the
apple
equally!



The guys
are
Thirsty!

Problem Statement General Formula Samples



Formula-1

Because of **[this technical problem]** existence, **[this real-life symptom/problem]** occurs. **[An illustrative text to explain the link between the Technical and the real-life sides of the problem WITH evidence(s) [reference] if available.]**

Formula-2

According to **[a reference]**, **[this real-life symptom/problem]** would appear due to **[this technical problem]**. **[An illustrative text to explain the link between the Technical and the real-life sides of the problem.]**

Problem Statement

Examples

∞ Example-1: Because of **[Covid-19 Virus occurrence]** in November 2019, **[all the students were forced to stay home as their schools were closed]**. **[upon the arrival of the Covid-19 Virus into the human's lungs, it would attack its healthy cells. The cells might become inflamed, making it tough for the human to breathe. This can lead to pneumonia, an infection of the tiny air sacs (called alveoli) inside the lungs where the blood exchanges oxygen and carbon dioxide. Eventually, if not treated in time, the human being might die.[WHO 2020]]**

Problem Statement

Examples

- ❧ Example-2: According to [NASA 2021], [the earth's temperature might increase few degrees next summer] due to [the detected rapid increment in air pollution for the last few months]. [It is well known that the higher the air pollution is, the higher carbon dioxide rate would be in the air. One of the most remarkable aspects of the paleoclimate record is the strong correspondence between temperature and the concentration of carbon dioxide in the atmosphere observed during the glacial cycles of the past several hundred thousand years. When the carbon dioxide concentration goes up, temperature goes up. When the carbon dioxide concentration goes down, temperature goes down.]

Problem Statement

Examples

❧ **Example-3:** The rapid growth of online content has made it increasingly challenging for users to find relevant and accurate information efficiently. Despite advancements in search algorithms, users still struggle to obtain precise results tailored to their needs. Traditional keyword-based search engines often return irrelevant or outdated content, leading to user frustration and wasted time.

Problem Statement

Examples

☞ **Example-4:** The problem of overfitting in machine learning models poses a critical challenge, particularly in scenarios with limited labeled data or high-dimensional feature spaces. Overfitting occurs when a model captures noise or random fluctuations in the training data, leading to poor generalization performance on unseen data. This phenomenon is exacerbated in complex models such as deep neural networks, where millions of parameters can easily memorize training examples. Addressing overfitting is essential for building robust and reliable machine learning systems that can accurately generalize to new data.



Developing Good Research Objectives

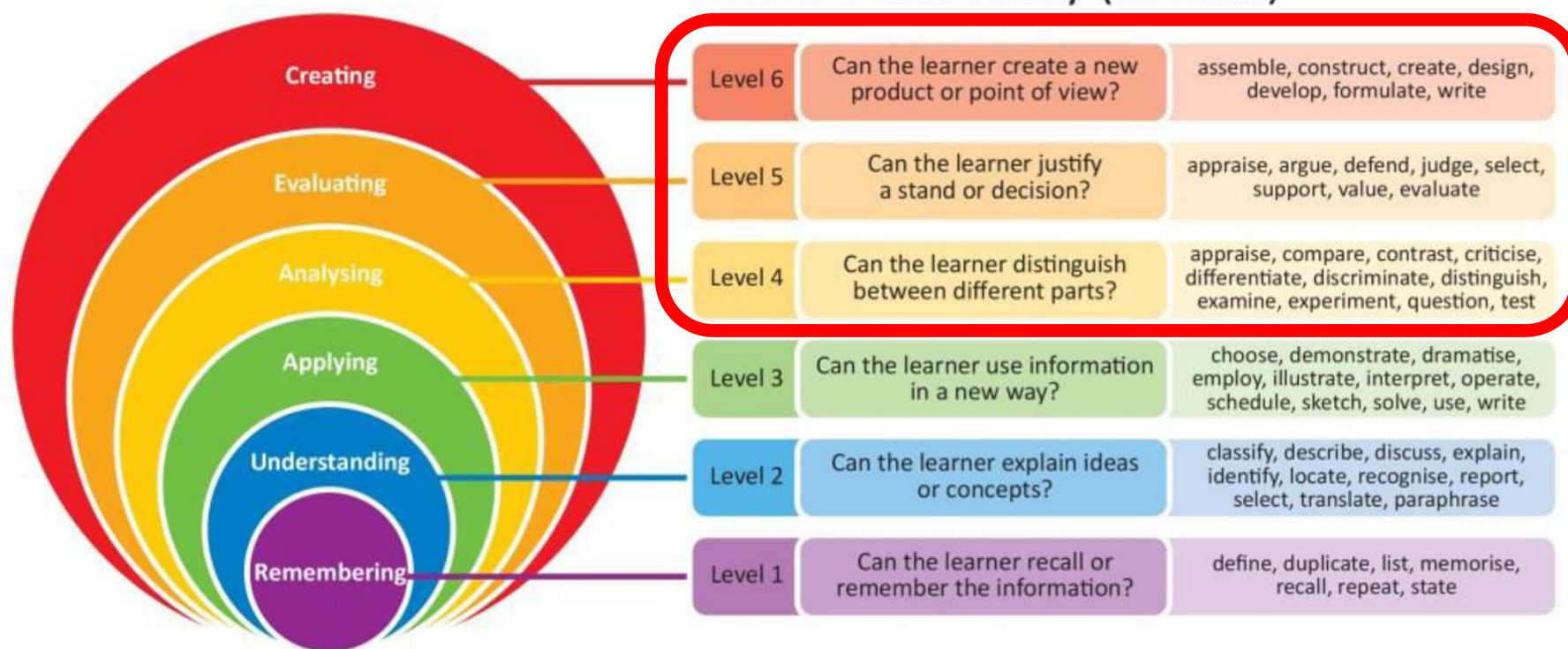
Developing Good Research Objectives

1. They should be presented briefly and concisely
2. They should be presented in logical sequence
3. They should be realistic (e.g., achieved within the expected timeframe, achieved within the available resources)
4. They should use action verbs that are specific enough to be evaluated or measured (e.g., assess, determine, compare, verify, calculate, describe)
5. They must be measurable.
6. They should be static once the study work begins (i.e., objectives should not be moving targets)

Objectives' Verbs



Bloom's taxonomy (revised) **Preferrable**



Research Objectives Sample



Research Objectives

- 1.To investigate the short-term and long-term impacts of COVID-19 on edu-tourism industry.
- 2.To determine the factors that drive students' better satisfaction in the hybrid and online learning environment.
- 3.To identify the innovative instructional strategies (i.e develop a software or program) in engaging students during the hybrid and online classes with various contexts using web and mobile technologies.
- 4.To identify the advantages, disadvantages, constraints, and benefits of hybrid and online learning activities.

Missing 'To evaluate...'

Research Objectives Sample



Research Objectives

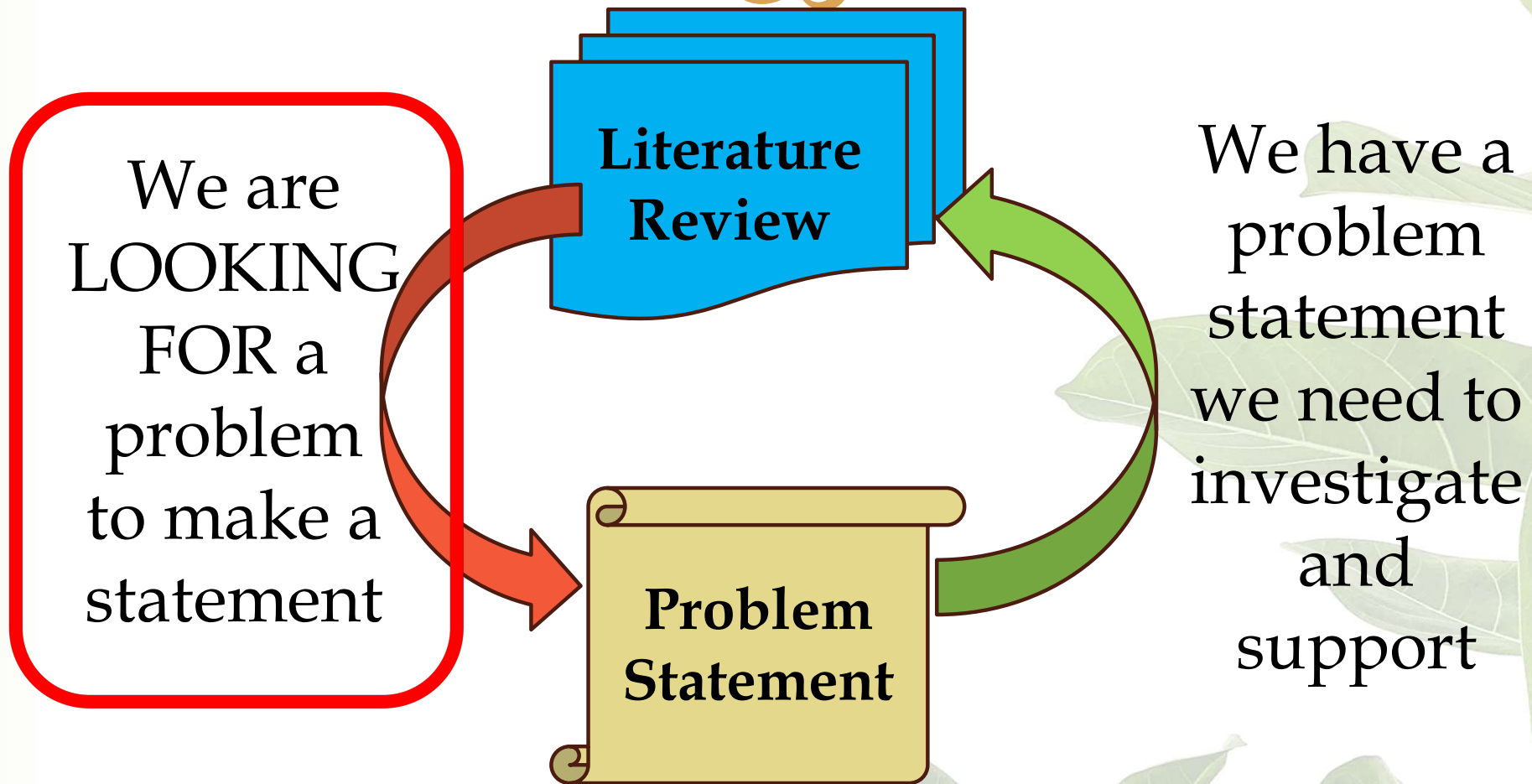
1. To design and develop a framework for managing a medium-size enterprise company during a worldwide economical crises.
2. To evaluate the developed framework via trial deployment in three medium-size enterprises.
 - a) To evaluate the framework's fitness.
 - b) To evaluate the framework's scalability.

} **Sub Objectives**

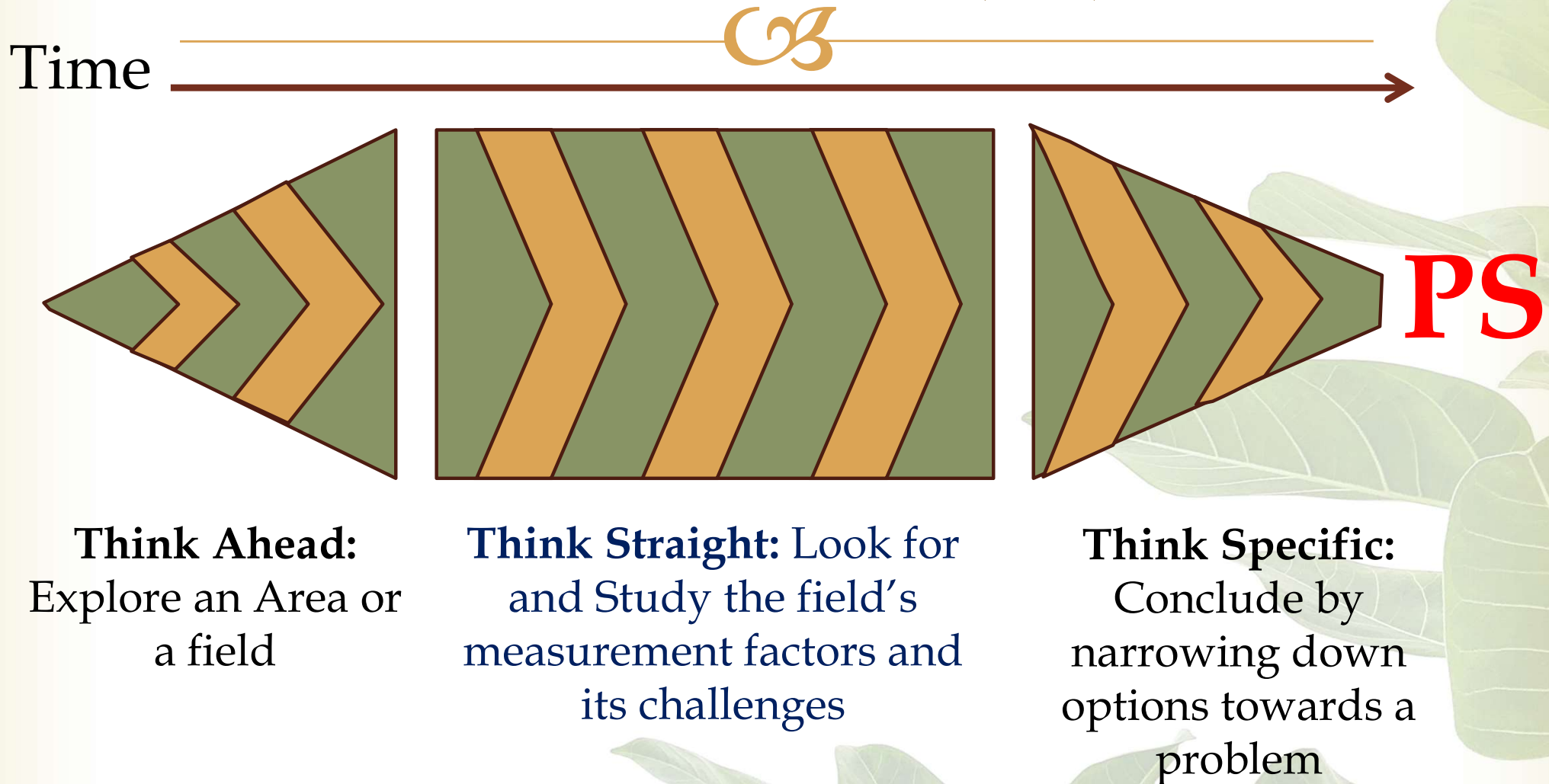


Conducting Literature Review

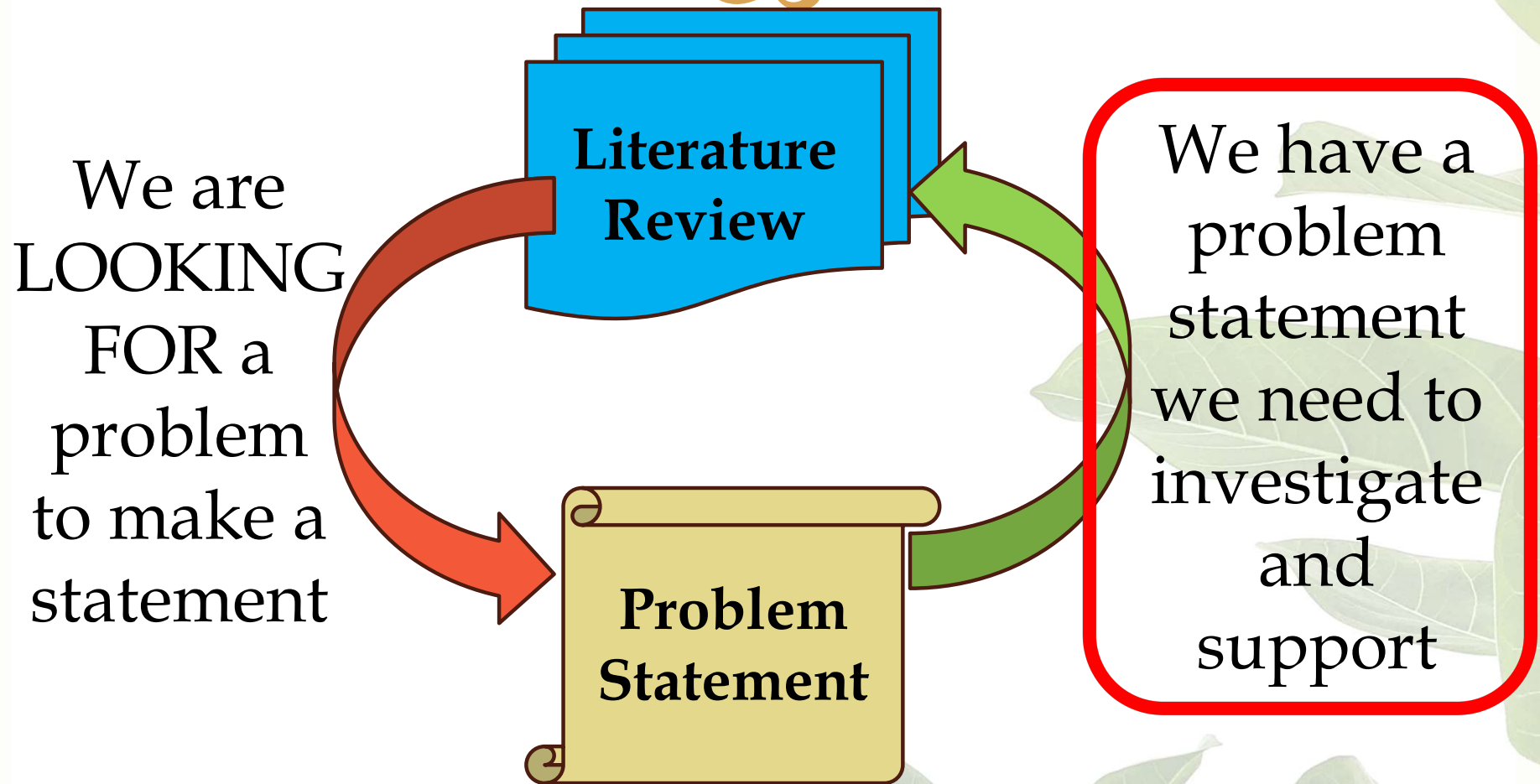
Literature Review



From Literature Review to Problem Statement (PS)



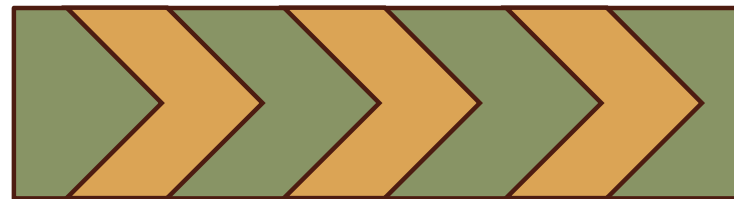
Literature Review



Support a Problem Statement by Literature Review

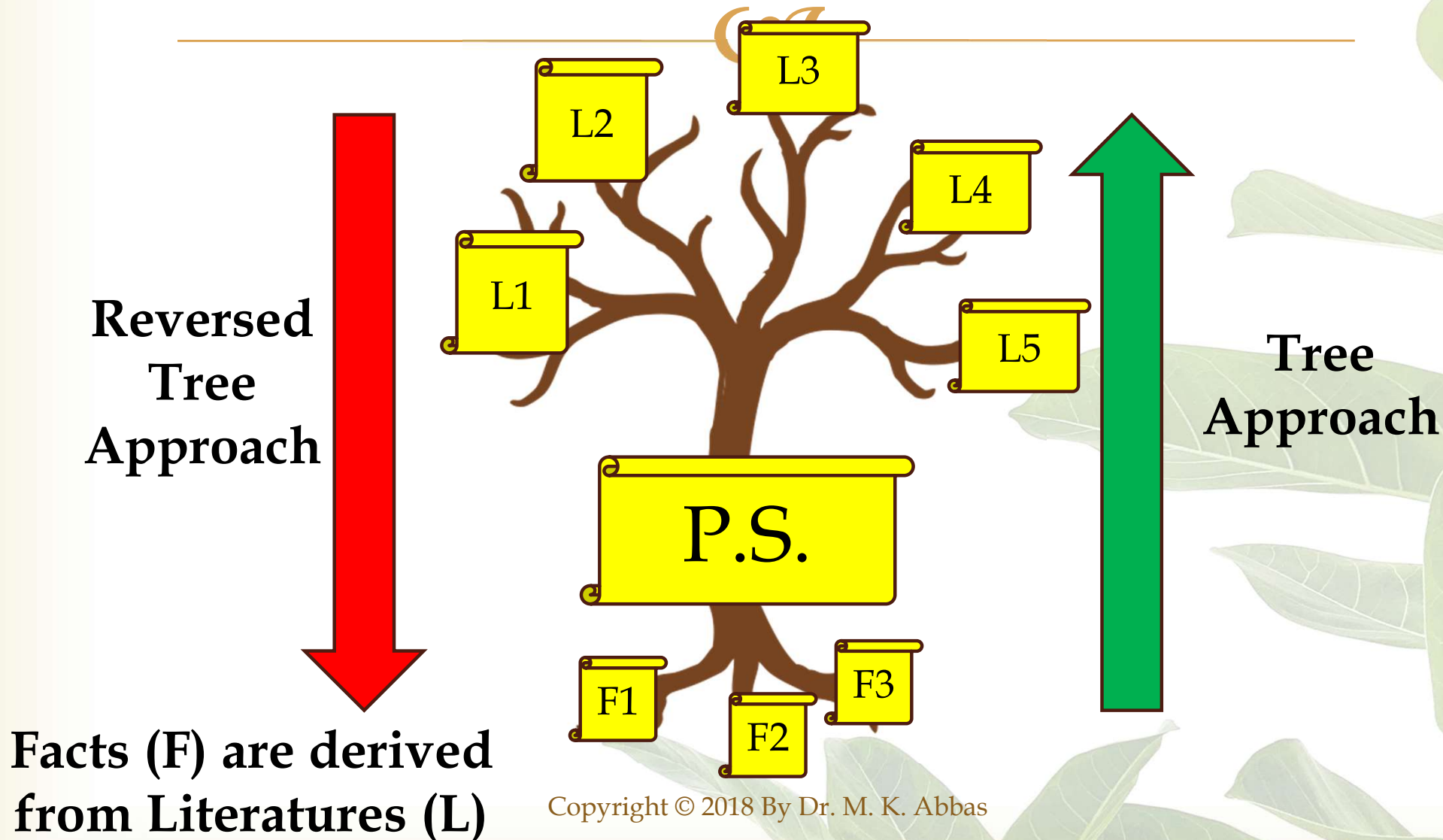


Time 



**Think Precise: Critically
criticize other
methodologies to support
your Problem Statement**

Writing the Problem Statement from Literature Review



ABSTRACTS



❧ What is an Abstract?

❧ A brief summary of a research article (Thesis, journal paper, or a conference paper). Abstracts help the readers to grasp fast the main context that article is talking about.

ABSTRACTS



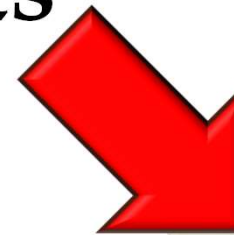
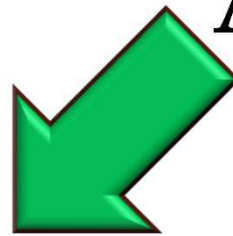
- ❧ Why we need to write an Abstract?
 - ❧ Convince the reader to read the rest of the paper.
 - ❧ Or even seduce the reader to purchase the paper.

- ❧ What are the main features of abstracts?
 - ❧ Summarizes all the parts of the article
 - ❧ Must have the most important parts
 - ❧ Accurate
 - ❧ Concise
 - ❧ Captivating (grasps the reader's attention)

Writing an Abstract



The Two types of Abstracts



Descriptive

Informative



Structured

Unstructured



Structured

Unstructured

Descriptive Abstract



- It is a single paragraph of ~100 -200 words.
- It consists of three parts
 - ❧ Purpose
 - ❧ Objective
 - ❧ Methods
 - ~~❧ Results~~
 - ~~❧ Conclusion~~

Informative Abstract



- It is the common type used for conferences and journal papers.
- It consists of **FIVE** parts
 - ❧ Purpose and Motivation
 - ❧ Problem
 - ❧ Methods
 - ❧ Results
 - ❧ Conclusion

Structured Abstract



❧ Structured abstracts have explicit headings showing what section of the abstract we are writing.

❧ Example: **Alcohol intake and risk of incident gout in men: a prospective study**
Hyon K Choi, Karen Atkinson, Elizabeth W Karlson, Walter Willett, Gary Curhan

Summary

Background The association between alcohol consumption and risk of gout has been suspected since ancient times, but has not been prospectively confirmed. Additionally, potential differences in risk of gout posed by different alcoholic beverages have not been assessed.

Methods Over 12 years (1986–98) we used biennial questionnaires to investigate the relation between alcohol consumption and risk of incident gout in 47 150 male participants with no history of gout at baseline. We used a supplementary questionnaire to ascertain whether reported cases of gout met the American College of Rheumatology survey gout criteria.

Findings We documented 730 confirmed incident cases of gout. Compared with men who did not drink alcohol, the multivariate relative risk (RR) of gout was 1.32 (95% CI 0.99–1.75) for alcohol consumption 10.0–14.9 g/day, 1.49 (1.14–1.94) for 15.0–29.9 g/day, 1.96 (1.48–2.60) for 30.0–49.9 g/day, and 2.52 (1.72–3.70) for ≥50 g/day (p-trend <0.0001). Beer consumption showed the

Unstructured Abstract



✧ Unstructured abstracts are just one or more text paragraphs.

✧ Example:

Abstract

Many forms of inductive logic programming (ILP) use *metarules*, second-order Horn clauses, to define the structure of learnable programs and thus the hypothesis space. Deciding which metarules to use for a given learning task is a major open problem and is a trade-off between efficiency and expressivity: the hypothesis space grows given more metarules, so we wish to use fewer metarules, but if we use too few metarules then we lose expressivity. In this paper, we study whether fragments of metarules can be logically reduced to minimal finite subsets. We consider two traditional forms of logical reduction: subsumption and entailment. We also consider a new reduction technique called *derivation reduction*, which is based on SLD-resolution. We compute reduced sets of metarules for fragments relevant to ILP and theoretically show whether these reduced sets are reductions for more general infinite fragments. We experimentally compare learning with reduced sets of metarules on three domains: Michalski trains, string transformations, and game rules. In general, derivation reduced sets of metarules outperform subsumption and entailment reduced sets, both in terms of predictive accuracies and learning times.

The most common way for write Abstract

Abstract is ONE SINGLE PARAGRAPH that has three elements about the research:-

- ❧ Problem Statement
- ❧ Introduced Solution
- ❧ (Expected) Outcomes / Findings

Writing a Proposal



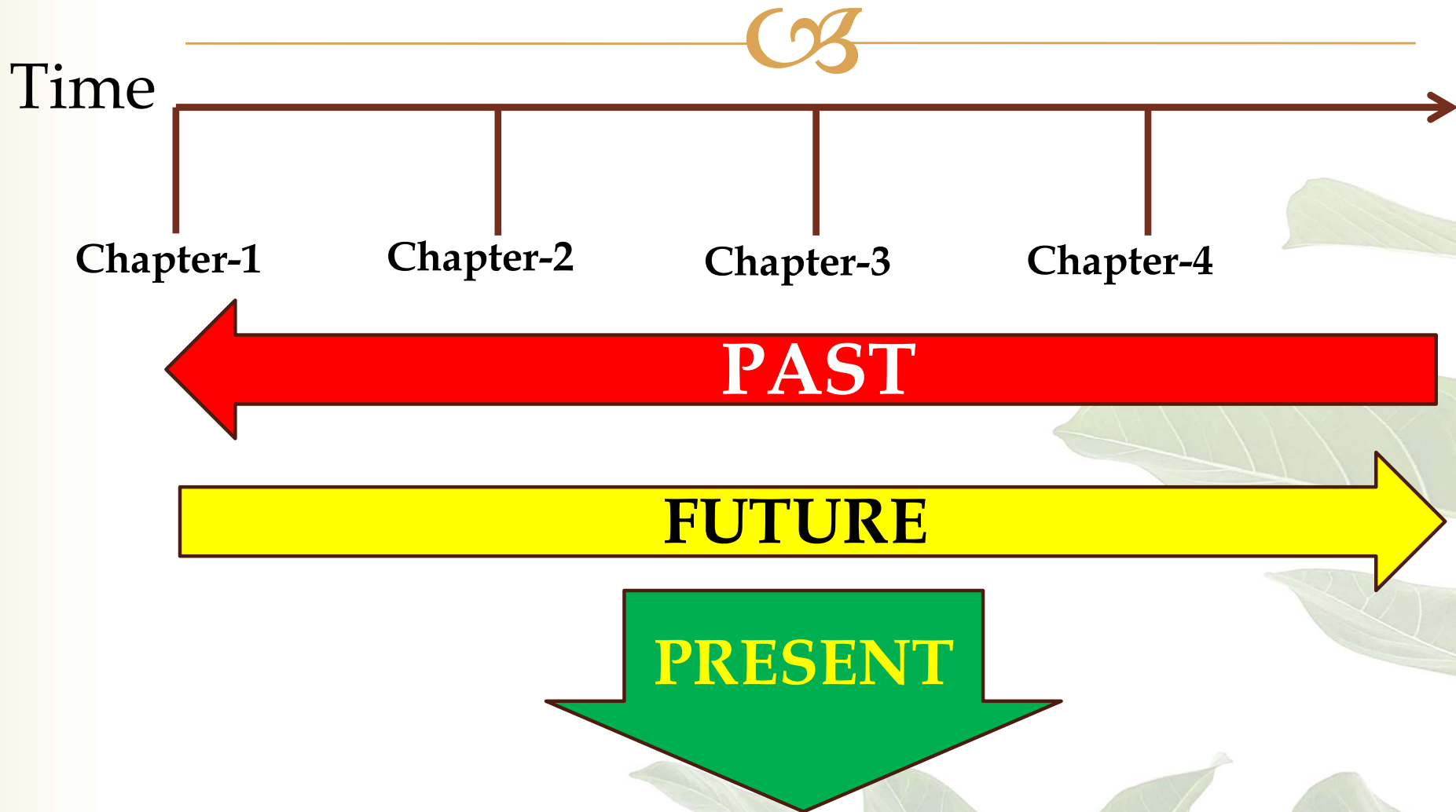
- ❧ Abstract
- ❧ 1. Introduction
- ❧ 2. Background
- ❧ 3. Problem Statement(s)
- ❧ 4. Aim(s)
- ❧ 5. Objective(s)
- ❧ 6. Research Question(s)
- ❧ 7. Research Scope
- ❧ 8. Related Work / Literature Review
- ❧ 9. Methodology
- ❧ 10. Conclusion
- ❧ Gantt Chart
- ❧ References

Writing a Dissertation



- ❧ Abstract
- ❧ Chapter-1: Introduction
 - ❧ Overview
 - ❧ Background
 - ❧ Problem Statement(s)
 - ❧ Aim(s)
 - ❧ Objective(s)
 - ❧ Research Question(s)
 - ❧ Research Scope
 - ❧ Thesis Layout
- ❧ Chapter-2: Literature Review
- ❧ Chapter-3: Methodology
- ❧ Chapter-4: Results and Discussion
- ❧ Chapter-5: Conclusion and Future Recommendations
- ❧ References

Chapters Language





End of Part-2